DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

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MITRE ATT&CK vs Cyber Kill Chain

Learning Objective

To understand the difference between MITRE ATT&CK and the Cyber Kill Chain and learn how both can be used during security investigations.

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1. What is MITRE ATT&CK?

MITRE ATT&CK is a knowledge base that focuses on **how attackers behave** during a cyber attack.

It contains different tactics and techniques that attackers use in real-world attacks.

For example:

- Phishing
- PowerShell
- Credential Dumping
- Network Service Scanning
- Remote Services

A SOC analyst can use ATT&CK to investigate alerts and understand the techniques being used by an attacker.

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2. What is the Cyber Kill Chain?

The **Cyber Kill Chain** is a model created by Lockheed Martin that describes the different stages of a cyber attack.

It has **7 stages**:

1. Reconnaissance
2. Weaponization

3. Delivery
4. Exploitation
5. Installation
6. Command and Control
7. Actions on Objectives

It helps security teams understand an attack as a sequence of stages.

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3. Difference between MITRE ATT&CK and Cyber Kill Chain

MITRE ATT&CK	Cyber Kill Chain
Focuses on attacker behaviour	Focuses on stages of an attack
Contains tactics and techniques	Contains 7 attack stages
More detailed	More high-level
Based on real-world attacker behaviour	Provides a general attack lifecycle
Useful for threat hunting and detection	Useful for understanding the overall attack process
Can contain many techniques for one tactic	Has a fixed sequence of stages
Commonly used by SOC and threat detection teams	Commonly used for understanding and stopping attacks at different stages
Example: PowerShell technique	Example: Exploitation stage

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4. Simple Example

Suppose an attacker sends a malicious email to an employee.

Using the **Cyber Kill Chain**, we can look at the attack as part of the delivery stage.

Using **MITRE ATT&CK**, we can identify the specific behaviour as:

Tactic: Initial Access
Technique: Phishing

If the attacker later uses PowerShell, ATT&CK can also help us identify:

Tactic: Execution
Technique: Command and Scripting Interpreter: PowerShell

So, ATT&CK gives more detail about the specific behaviour.

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5. Which one is more useful for a SOC analyst?

I think both are useful, but they are useful in different ways.

Cyber Kill Chain

It helps me understand:

- Where the attack is in its overall lifecycle
- What the attacker may do next
- Where security controls can stop the attack

MITRE ATT&CK

It helps me understand:

- What technique the attacker is using
- What logs I should investigate
- What detection rules could be created
- What other attacker behaviour may be related

For day-to-day SOC investigations, I would expect MITRE ATT&CK to be more useful because it provides more detailed information about attacker behaviour.

6. SOC Investigation Workflow

A SOC analyst can use both MITRE ATT&CK and the Cyber Kill Chain while investigating a suspicious alert.

A basic workflow can be:

Alert Received
↓
Check Alert Details
↓
Collect Related Logs / Evidence
↓
Understand the Activity
↓
Identify Attack Stage
↓
Map Behaviour to MITRE ATT&CK
↓
Check for Related Techniques / Events
↓
Decide: Benign or Suspicious
↓
Respond / Escalate
↓
Document the Investigation

Example

If I receive an alert for a suspicious PowerShell command, I would:

1. Check the alert and timestamp.
2. Identify the affected system and user.
3. Check the PowerShell command that was executed.
4. Look at the parent process and other related processes.
5. Check network connections made by the process.

6. Look for other suspicious events around the same time.
7. Map the behaviour to the relevant MITRE ATT&CK technique.
8. Consider where this activity fits into the overall attack.
9. Decide whether it looks legitimate or suspicious.
10. Escalate or take the required response action if needed.
11. Document the findings.

This workflow helps me avoid making a decision based on only one alert and instead look at the complete activity around it.

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7. My Understanding

The main difference I understood is that the **Cyber Kill Chain gives a high-level view of an attack**, while **MITRE ATT&CK gives more detailed information about the techniques attackers use**.

For example, the Kill Chain can help me understand the overall stages of an attack, while ATT&CK can help me identify that the attacker used something like phishing or PowerShell.

So I don't think they are replacements for each other. They can be used together to get a better understanding of an attack.